SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 0603 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptron’s, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS:15 **Total Marks:25**

Annexure A

DEBUGGING & OPTIMIZATION TASK QUESTIONS

Code of Conduct:

I X. Joshua – (192312188) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

X. Joshua

1. Perceptron Model Fails to Converge

Answer

A perceptron should converge if the dataset is linearly separable. Failure to converge may occur due to:

- **Improper learning rate:** A very high learning rate causes oscillations, while a very low rate slows learning.
- **Incorrect weight initialization:** Extremely large initial weights can delay convergence.
- **Bias handling errors:** Missing or incorrect bias prevents finding the correct decision boundary.
- **Incorrect update rule:** Errors in updating weights or labels stop proper learning.

Debugging Steps

1. Verify that the dataset is linearly separable.
2. Check the perceptron update equation.
3. Ensure bias is included.
4. Monitor weight updates after every iteration.
5. Test using a small sample dataset.

Optimization Techniques

- Use a moderate learning rate.
- Initialize weights with small random values.
- Normalize input features.
- Shuffle training samples before each epoch.
- Stop training when no classification errors remain.

2. Slow Convergence in Backpropagation

Answer

Slow convergence and local minima occur because of:

- Vanishing gradients
- Poor activation functions
- Improper weight initialization
- Incorrect learning rate

Debugging Steps

1. Monitor loss after each epoch.
2. Check gradients for very small values.
3. Verify forward and backward propagation.
4. Inspect weight updates.
5. Compare training and validation accuracy.

Optimization Techniques

- Use ReLU instead of sigmoid where appropriate.
- Apply Xavier or He initialization.
- Normalize input data.
- Use Adam optimizer.
- Apply adaptive learning rates and batch normalization.

3. Genetic Algorithm Produces Poor Solutions

Answer

Possible causes include:

- Incorrect chromosome encoding
- Poor fitness function
- Premature convergence
- Low population diversity

Debugging Steps

1. Verify chromosome representation.
2. Test the fitness function independently.
3. Monitor population diversity.
4. Check crossover and mutation operations.
5. Observe best fitness over generations.

Optimization Techniques

- Increase mutation rate slightly.
- Adjust selection pressure.
- Apply elitism.
- Increase population size.

4. Deep Neural Network with Genetic Programming

Answer

The model suffers from overfitting and high computational cost because of:

- Excessive network complexity
- Too many parameters
- Limited training data
- Inefficient evaluation

Debugging Steps

1. Compare training and validation accuracy.
2. Monitor loss curves.
3. Identify unnecessary layers.
4. Measure computation time.
5. Check data quality.

Optimization Techniques

- Apply L1/L2 regularization.
- Use dropout.
- Prune unnecessary neurons.
- Apply early stopping.
- Reduce model complexity.

5. Decision Tree Overfitting

Answer

High training accuracy but poor testing accuracy indicates overfitting.

Possible Causes

- Deep tree
- No pruning
- Noisy features
- Small training dataset

Debugging Steps

1. Compare training and testing accuracy.
2. Check tree depth.

3. Analyze feature importance.
4. Remove noisy features.
5. Perform cross-validation.

Optimization Techniques

- Pre-pruning
- Post-pruning
- Feature selection
- Cross-validation
- Increase training data

6. CNN Produces Unstable Results

Answer

Unstable CNN training may result from:

- Improper data augmentation
- Vanishing gradients
- Batch normalization issues
- High learning rate

Debugging Steps

1. Check training loss.
2. Verify augmented images.
3. Inspect gradient values.
4. Validate batch normalization.
5. Monitor learning rate.

Optimization Techniques

- Use Batch Normalization.
- Reduce learning rate.
- Apply data normalization.
- Use ReLU activation.
- Employ learning rate scheduling.

7. Reinforcement Learning Agent Fails

Answer

Possible causes:

- Poor reward design
- Insufficient exploration
- Incorrect state representation
- Poor hyperparameters

Debugging Steps

1. Examine reward signals.
2. Observe agent actions.
3. Verify state representation.
4. Monitor cumulative rewards.
5. Test different exploration rates.

Optimization Techniques

- Reward shaping
- ε-greedy exploration
- Hyperparameter tuning
- Better state representation
- Experience replay

8. K-Means Gives Inconsistent Clusters

Answer

Possible causes:

- Random centroid initialization
- Wrong K value
- Outliers
- Different feature scales

Debugging Steps

1. Check initial centroids.
2. Normalize features.
3. Remove outliers.

4. Evaluate cluster quality.
5. Repeat experiments.

Optimization Techniques

- K-Means++
- Elbow Method
- Silhouette Analysis
- Feature scaling
- Outlier removal

9. SVM Performs Poorly on Nonlinear Data

Answer

Possible causes:

- Incorrect kernel
- Improper C and gamma values
- Feature scaling problems

Debugging Steps

1. Verify feature scaling.
2. Test different kernels.
3. Compare model accuracy.
4. Analyze confusion matrix.
5. Tune hyperparameters.

Optimization Techniques

- RBF kernel
- Grid Search
- Feature normalization
- Cross-validation
- Parameter tuning

10. RNN Cannot Learn Long-Term Dependencies

Answer

Possible causes:

- Vanishing gradients

- Limited memory
- Long sequences

Debugging Steps

1. Monitor training loss.
2. Check gradient values.
3. Analyze sequence length.
4. Inspect hidden states.
5. Compare training performance.

Optimization Techniques

- Use LSTM
- Use GRU
- Apply Attention Mechanism
- Gradient clipping
- Better learning rate scheduling

11. SVM Shows Low Training and Testing Accuracy

Answer

Low accuracy on both datasets indicates underfitting.

Possible Causes

- Incorrect kernel
- Poor hyperparameter tuning
- Unscaled features

Debugging Steps

1. Normalize data.
2. Test Linear, Polynomial, and RBF kernels.
3. Tune C and gamma.
4. Evaluate confusion matrix.
5. Perform cross-validation.

Optimization Techniques

- Grid Search
- Feature scaling

- Hyperparameter tuning
- Kernel optimization
- Feature engineering

12. K-Means Forms Imbalanced Clusters

Answer

Possible causes:

- Wrong K
- Random centroid initialization
- Outliers

Debugging Steps

1. Check cluster sizes.
2. Plot data distribution.
3. Detect outliers.
4. Test multiple K values.
5. Compare clustering metrics.

Optimization Techniques

- Elbow Method
- K-Means++
- Remove outliers
- Normalize data
- Silhouette Analysis

13. RNN Gives Inconsistent Time-Series Predictions

Answer

Possible causes:

- Vanishing gradients
- Short input sequences
- Poor architecture

Debugging Steps

1. Monitor prediction errors.
2. Inspect gradient values.

3. Test sequence length.
4. Validate preprocessing.
5. Compare different architectures.

Optimization Techniques

- LSTM
- GRU
- Gradient clipping
- Sequence normalization
- Hyperparameter tuning

14. Naïve Bayes Performs Poorly on Text Data

Answer

Possible causes:

- Feature independence assumption violated
- Noisy features
- Poor preprocessing
- Imbalanced dataset

Debugging Steps

1. Review text preprocessing.
2. Remove stop words.
3. Check class distribution.
4. Evaluate confusion matrix.
5. Analyze important features.

Optimization Techniques

- TF-IDF feature selection
- Laplace smoothing
- Data balancing
- Remove irrelevant features
- Better text preprocessing

15. Reinforcement Learning Navigation Problem

Answer

Possible causes:

- Sparse rewards
- Poor exploration
- Incorrect reward function
- Improper hyperparameters

Debugging Steps

1. Observe agent movement.
2. Monitor reward values.
3. Test exploration rate.
4. Verify environment dynamics.
5. Analyze learning curves.

Optimization Techniques

- Reward shaping
- Epsilon decay tuning
- Policy optimization algorithms
- Hyperparameter tuning
- Experience replay